

✓ REMORSE — Exploratory Data Analysis

Reptilian Evaluation of Mesozoic Origins: Retrospective Study on Extinction

Questo notebook esplora i due dataset fossili della Paleobiology Database (PBDB):

- `dinos.csv` — occorrenze fossili di Dinosauria
- `plants.csv` — occorrenze fossili di Plantae (Mesozoico)

Obiettivo: capire la qualità dei dati, identificare anomalie e prendere decisioni informate sulle trasformazioni da applicare nella pipeline ETL.

✓ 0. Setup

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

pd.set_option('display.max_columns', None)
pd.set_option('display.float_format', '{:.2f}'.format)

print('Librerie caricate.')
```

Librerie caricate.

✓ 1. Caricamento dei dataset

```
dinos = pd.read_csv('data/raw/dinos.csv', low_memory=False)
plants = pd.read_csv('data/raw/plants.csv', low_memory=False)

print(f'dinos.csv → {dinos.shape[0]:,} righe, {dinos.shape[1]} colonne')
print(f'plants.csv → {plants.shape[0]:,} righe, {plants.shape[1]} colonne')
```

dinos.csv → 37,790 righe, 38 colonne
plants.csv → 57,394 righe, 38 colonne

✓ 2. Struttura dei dataset

```
print('=== DINOS ===')
dinos.info()
```

```
=== DINOS ===
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 37790 entries, 0 to 37789
Data columns (total 38 columns):
 #   Column                Non-Null Count  Dtype
---  -
 0   occurrence_no         37790 non-null  int64
 1   record_type           37790 non-null  object
 2   reid_no               4154 non-null   float64
 3   flags                 3417 non-null   object
 4   collection_no        37790 non-null  int64
 5   identified_name       37790 non-null  object
 6   identified_rank       37790 non-null  object
 7   identified_no         12703 non-null  float64
 8   difference            3889 non-null   object
 9   accepted_name        37790 non-null  object
10   accepted_attr        37748 non-null  object
11   accepted_rank        37790 non-null  object
12   accepted_no          37790 non-null  int64
13   early_interval       37790 non-null  object
14   late_interval        8675 non-null   object
15   max_ma               37790 non-null  float64
16   min_ma               37790 non-null  float64
17   reference_no         37790 non-null  int64
18   lng                  37790 non-null  float64
19   lat                  37790 non-null  float64
20   cc                   37773 non-null  object
21   state                31470 non-null  object
22   county              18729 non-null  object
23   latlng_basis         36179 non-null  object
24   latlng_precision    37790 non-null  object
25   altitude_value       8670 non-null   float64
26   altitude_unit        8579 non-null   object
27   geogscale            31163 non-null  object
```

```

28 geogcomments      29907 non-null object
29 cx_int_no          37728 non-null float64
30 formation          23443 non-null object
31 geological_group   10352 non-null object
32 member             7208 non-null object
33 phylum           37790 non-null object
34 class              37790 non-null object
35 order              30411 non-null object
36 family             29067 non-null object
37 genus              23558 non-null object
dtypes: float64(8), int64(4), object(26)
memory usage: 11.0+ MB

```

```

print('=== PLANTS ===')
plants.info()

```

```

=== PLANTS ===
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 57394 entries, 0 to 57393
Data columns (total 38 columns):
#   Column                Non-Null Count  Dtype
---  -
0   occurrence_no          57394 non-null  int64
1   record_type            57394 non-null  object
2   reid_no                 167 non-null    float64
3   flags                  14503 non-null  object
4   collection_no          57394 non-null  int64
5   identified_name         57394 non-null  object
6   identified_rank         57394 non-null  object
7   identified_no           3597 non-null   float64
8   difference              11512 non-null  object
9   accepted_name          57394 non-null  object
10  accepted_attr           56253 non-null  object
11  accepted_rank           57394 non-null  object
12  accepted_no             57394 non-null  int64
13  early_interval          57394 non-null  object
14  late_interval           19454 non-null  object
15  max_ma                  57394 non-null  float64
16  min_ma                  57394 non-null  float64
17  reference_no            57394 non-null  int64
18  lng                     57394 non-null  float64
19  lat                     57394 non-null  float64
20  cc                      57394 non-null  object
21  state                   37147 non-null  object
22  county                  10406 non-null  object
23  latlng_basis            53242 non-null  object
24  latlng_precision        57394 non-null  object
25  altitude_value          5816 non-null   float64
26  altitude_unit           4537 non-null   object
27  geogscale               27212 non-null  object
28  geogcomments            28132 non-null  object
29  cx_int_no               57324 non-null  float64
30  formation               44451 non-null  object
31  geological_group        7916 non-null   object
32  member                  6257 non-null   object
33  phylum                 56912 non-null  object
34  class                   56294 non-null  object
35  order                   55547 non-null  object
36  family                   55184 non-null  object
37  genus                   54748 non-null  object
dtypes: float64(8), int64(4), object(26)
memory usage: 16.6+ MB

```

```

# Colonne presenti in entrambi i dataset
comuni = set(dinos.columns) & set(plants.columns)
solo_dinos = set(dinos.columns) - set(plants.columns)
solo_plants = set(plants.columns) - set(dinos.columns)

print(f'Colonne comuni:      {len(comuni)}')
print(f'Solo in dinos:        {len(solo_dinos)} → {solo_dinos}')
print(f'Solo in plants:        {len(solo_plants)} → {solo_plants}')

```

```

Colonne comuni:      38
Solo in dinos:        0 → set()
Solo in plants:       0 → set()

```

3. Analisi dei valori nulli

```

def null_report(df, name):
    null_pct = (df.isnull().sum() / len(df) * 100).sort_values(ascending=False)
    null_pct = null_pct[null_pct > 0]
    print(f'\n=== {name} - colonne con valori nulli ===')
    print(null_pct.to_string())

```

```
null_report(dinos, 'DINOS')
null_report(plants, 'PLANTS')
```

```
=== DINOS - colonne con valori nulli ===
flags                90.96
difference            89.71
reid_no              89.01
member               80.93
altitude_unit        77.30
altitude_value        77.06
late_interval         77.04
geological_group      72.61
identified_no         66.39
county               50.44
formation             37.97
genus                 37.66
family                23.08
geogcomments          20.86
order                 19.53
geogscale             17.54
state                 16.72
latlng_basis          4.26
cx_int_no             0.16
accepted_attr         0.11
cc                    0.04
```

```
=== PLANTS - colonne con valori nulli ===
reid_no              99.71
identified_no         93.73
altitude_unit         92.09
altitude_value        89.87
member               89.10
geological_group      86.21
county               81.87
difference            79.94
flags                74.73
late_interval         66.10
geogscale             52.59
geogcomments          50.98
state                 35.28
formation             22.55
latlng_basis          7.23
genus                 4.61
family                3.85
order                 3.22
accepted_attr         1.99
class                 1.92
phylum              0.84
cx_int_no             0.12
```

```
# Percentuale di record con coordinate valide
dinos_coord = dinos[['lat', 'lng']].dropna()
plants_coord = plants[['lat', 'lng']].dropna()

print(f'Dinos con coordinate valide: {len(dinos_coord):,} ({len(dinos_coord)/len(dinos)*100:.1f}%)')
print(f'Plants con coordinate valide: {len(plants_coord):,} ({len(plants_coord)/len(plants)*100:.1f}%)')
```

```
Dinos con coordinate valide: 37,790 (100.0%)
Plants con coordinate valide: 57,394 (100.0%)
```

4. Duplicati

```
print(f'Duplicati in dinos: {dinos.duplicated().sum()}')
print(f'Duplicati in plants: {plants.duplicated().sum()}')

# Verifica unicità di occurrence_no
print(f'\noccurrence_no univoci in dinos: {dinos["occurrence_no"].nunique():,} / {len(dinos):,}')
print(f'occurrence_no univoci in plants: {plants["occurrence_no"].nunique():,} / {len(plants):,}')
```

```
Duplicati in dinos: 0
Duplicati in plants: 0

occurrence_no univoci in dinos: 37,790 / 37,790
occurrence_no univoci in plants: 57,394 / 57,394
```

5. Distribuzione per periodo geologico

```
print('=== DINOS - early_interval (top 20) ===')
print(dinos['early_interval'].value_counts().head(20))
```

```

=== DINOS - early_interval (top 20) ===
early_interval
Holocene          5409
Late Pleistocene  4772
Late Campanian    2619
Late Maastrichtian 1511
Maastrichtian     1387
Kimmeridgian     1109
Middle Campanian  955
Aptian            955
Hettangian        800
Early Maastrichtian 696
Campanian         637
Lancian           589
Late Kimmeridgian 558
Middle Pleistocene 541
Norian            522
Late Barremian    490
Pleistocene       486
Early Cenomanian  471
Rancholabrean    437
Tithonian         435
Name: count, dtype: int64

```

```

print('=== PLANTS - early_interval (top 20) ===')
print(plants['early_interval'].value_counts().head(20))

```

```

=== PLANTS - early_interval (top 20) ===
early_interval
Late Jurassic      3395
Maastrichtian      3038
Albian             2846
Carnian            2510
Middle Jurassic    2284
Late Aptian        2238
Early Jurassic     2126
Aptian            1968
Norian            1957
Aalenian          1459
Early Albian       1445
Lancian           1421
Cenomanian        1398
Late Albian        1251
Bajocian          1202
Bathonian         1199
Barremian         1082
Late Triassic      1067
Late Maastrichtian 1060
Pliensbachian     1037
Name: count, dtype: int64

```

```

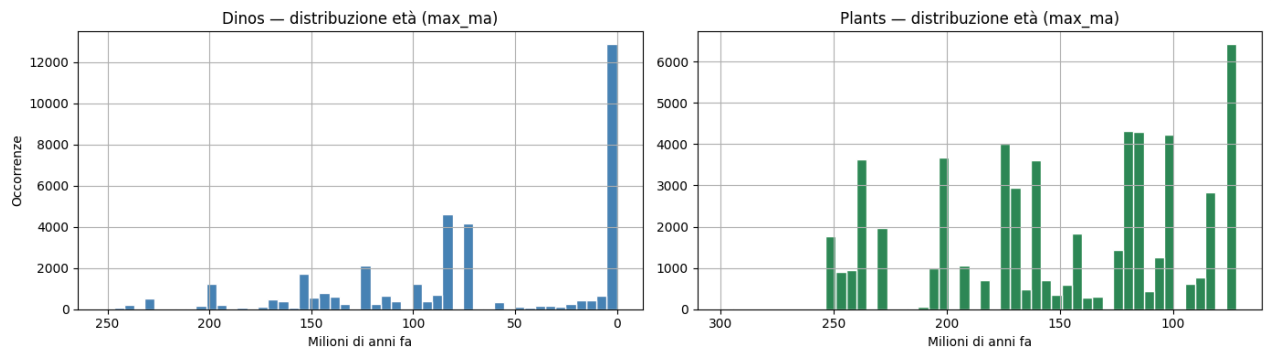
# Distribuzione per era (max_ma come proxy)
fig, axes = plt.subplots(1, 2, figsize=(14, 4))

dinos['max_ma'].dropna().hist(bins=50, ax=axes[0], color='steelblue', edgecolor='white')
axes[0].set_title('Dinos - distribuzione età (max_ma)')
axes[0].set_xlabel('Milioni di anni fa')
axes[0].set_ylabel('Occorrenze')
axes[0].invert_xaxis()

plants['max_ma'].dropna().hist(bins=50, ax=axes[1], color='seagreen', edgecolor='white')
axes[1].set_title('Plants - distribuzione età (max_ma)')
axes[1].set_xlabel('Milioni di anni fa')
axes[1].invert_xaxis()

plt.tight_layout()
plt.show()

```



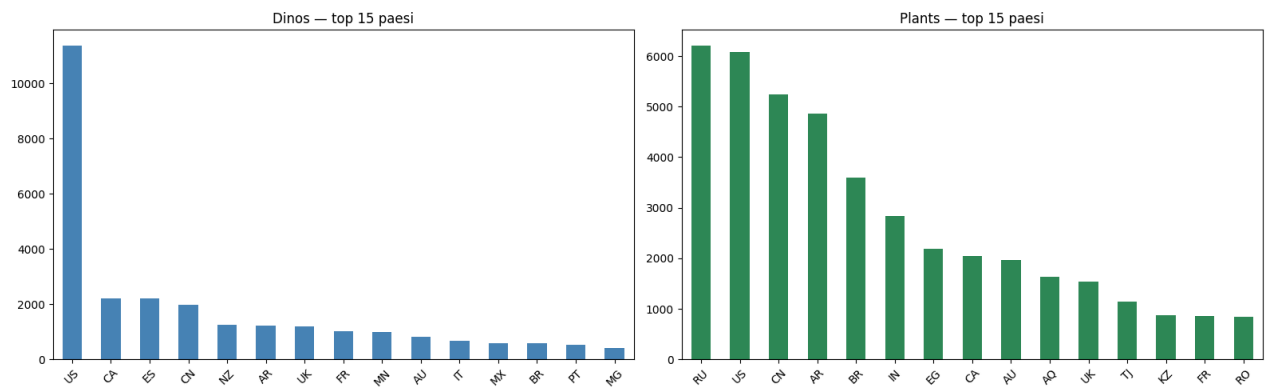
6. Distribuzione geografica

```
# Top 15 paesi per numero di ritrovamenti
fig, axes = plt.subplots(1, 2, figsize=(16, 5))

dinos['cc'].value_counts().head(15).plot(kind='bar', ax=axes[0], color='steelblue')
axes[0].set_title('Dinos — top 15 paesi')
axes[0].set_xlabel('')
axes[0].tick_params(axis='x', rotation=45)

plants['cc'].value_counts().head(15).plot(kind='bar', ax=axes[1], color='seagreen')
axes[1].set_title('Plants — top 15 paesi')
axes[1].set_xlabel('')
axes[1].tick_params(axis='x', rotation=45)

plt.tight_layout()
plt.show()
```



7. Classificazione tassonomica

```
print('=== DINOS — classi più frequenti ===')
print(dinos['class'].value_counts().head(10))

print('\n=== DINOS — Aves nel dataset ===')
aves = dinos[dinos['class'] == 'Aves']
print(f'Record Aves: {len(aves):,} ({len(aves)/len(dinos)*100:.1f}%)')

=== DINOS — classi più frequenti ===
class
```

```
Aves          15173
Reptilia      11164
Ornithischia  7058
Saurischia    4395
Name: count, dtype: int64

=== DINOS - Aves nel dataset ===
Record Aves: 15,173 (40.2%)
```

```
print('=== PLANTS - phylum più frequenti ===')
print(plants['phylum'].value_counts().head(10))
```

```
=== PLANTS - phylum più frequenti ===
phylum
Pteridophyta      16141
Gymnospermae      8378
Angiospermae      7308
Ginkgophyta       4648
Equisetophyta     3357
Cycadophyta       3009
NO_PHYLUM_SPECIFIED 2663
Chlorophyta       1787
Spermatophyta     1679
Lycopodiophyta    1487
Name: count, dtype: int64
```

8. Statistiche descrittive su max_ma / min_ma

```
for name, df in [('DINOS', dinos), ('PLANTS', plants)]:
    print(f'\n=== {name} ===')
    print(df[['max_ma', 'min_ma']].describe())
```

```
=== DINOS ===
      max_ma  min_ma
count 37790.00 37790.00
mean    71.01   64.52
std     65.82   61.62
min      0.01    0.00
25%     0.13    0.01
50%     72.20   66.00
75%    121.40  100.50
max     251.90  248.10

=== PLANTS ===
      max_ma  min_ma
count 57394.00 57394.00
mean   149.34  135.53
std     55.10   53.94
min     72.20    0.00
25%    100.50   93.90
50%    143.10  121.40
75%    192.90  174.70
max     298.90  250.50
```

9. Decisioni di trasformazione

Compilare questa sezione al termine dell'EDA con le decisioni prese dal gruppo.

Trasformazione	Decisione	Note
Escludere Aves	Da decidere	Vedere % in sezione 7
Gestione nulli	Da decidere	Vedere sezione 3
Eliminazione duplicati	Sì	Vedere sezione 4
Colonna <code>mid_ma</code>	Sì	$(\text{max_ma} + \text{min_ma}) / 2$
Colonna <code>period_group</code>	Sì	Triassic / Jurassic / Cretaceous
Colonna <code>dataset_type</code>	Sì	Dinosauria / Plantae
Flag <code>has_valid_coords</code>	Sì	Booleano su lat/long
Normalizzazione paesi	Da decidere	Vedere distribuzione in sezione 6